



SBER: Slot-Based Evidence Routing for Budgeted Multi-Agent Coordination

Yan Li

Beijing University of Posts and Telecommunications (BUPT)



► 1. Introduction

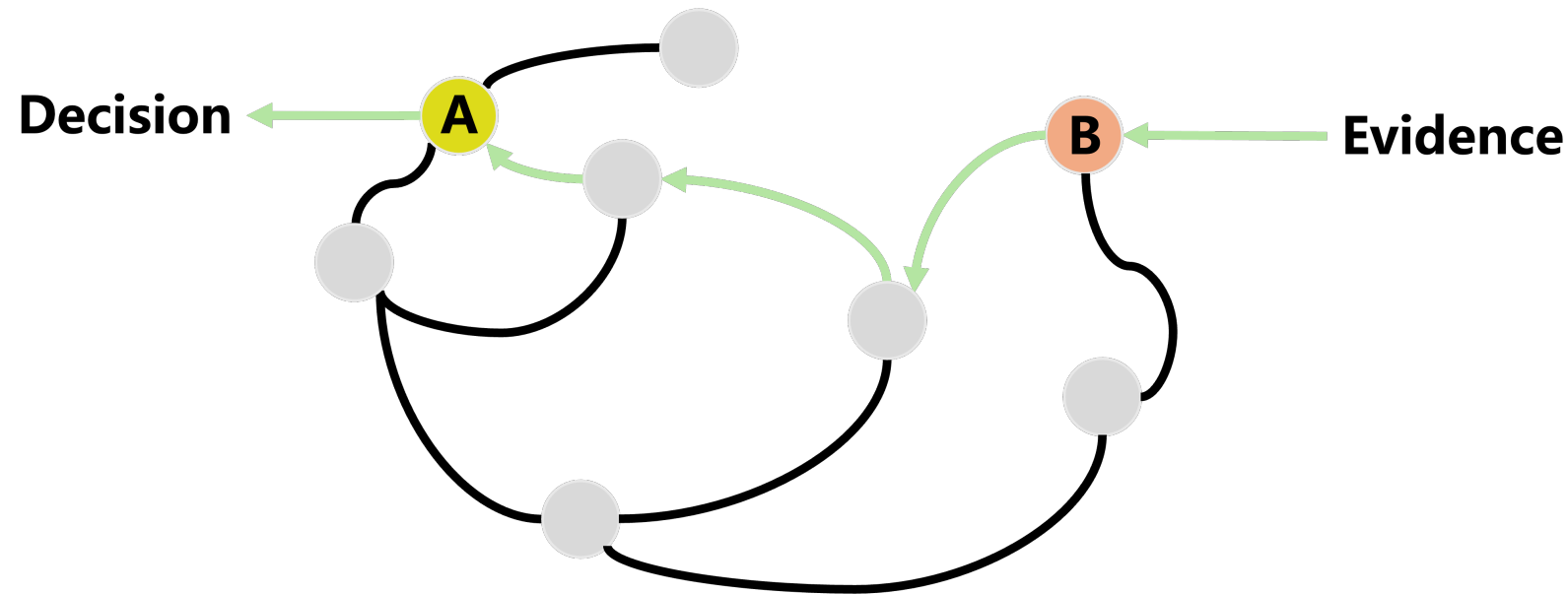
2. Methodology

3. Experiments

4. Conclusion

When the team sees the same world...

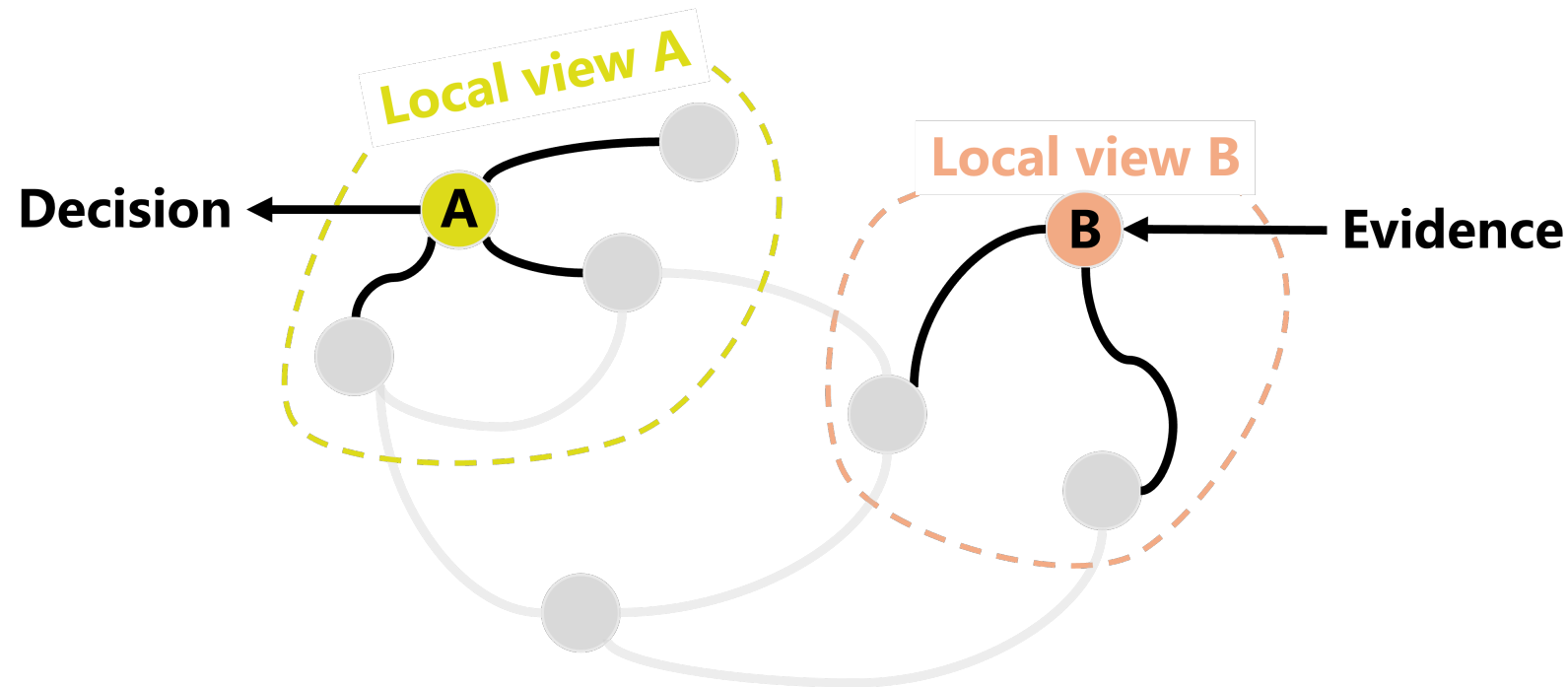
- *The agent or controller has access to the complete state;*
- *communication is more about synchronizing plans.*



Shared state → joint planning

When evidence is split across agents...

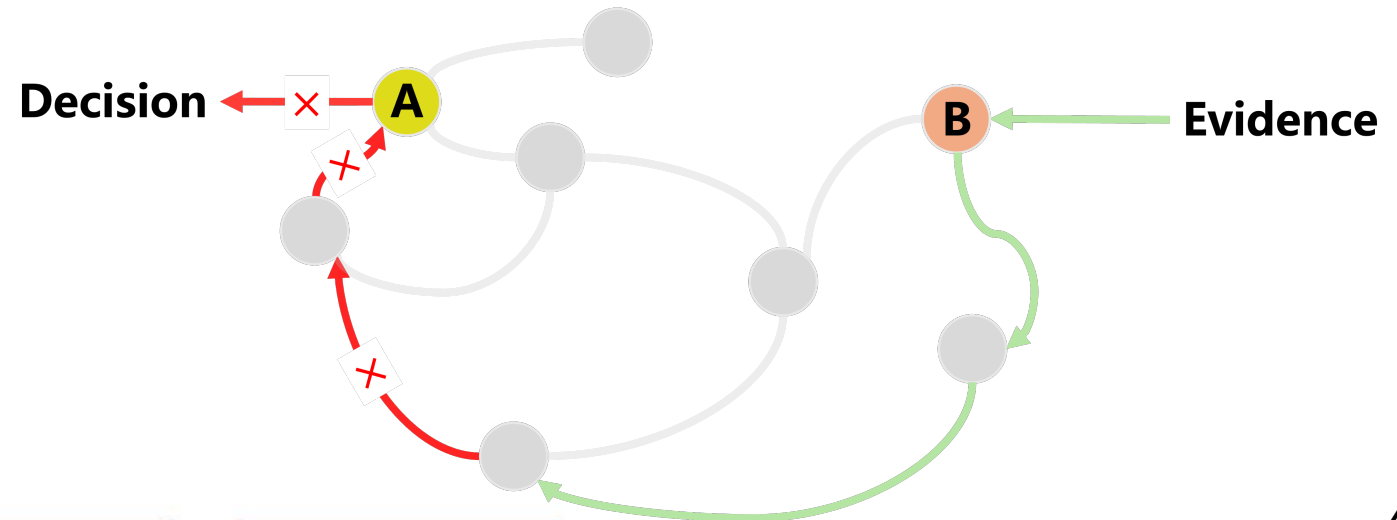
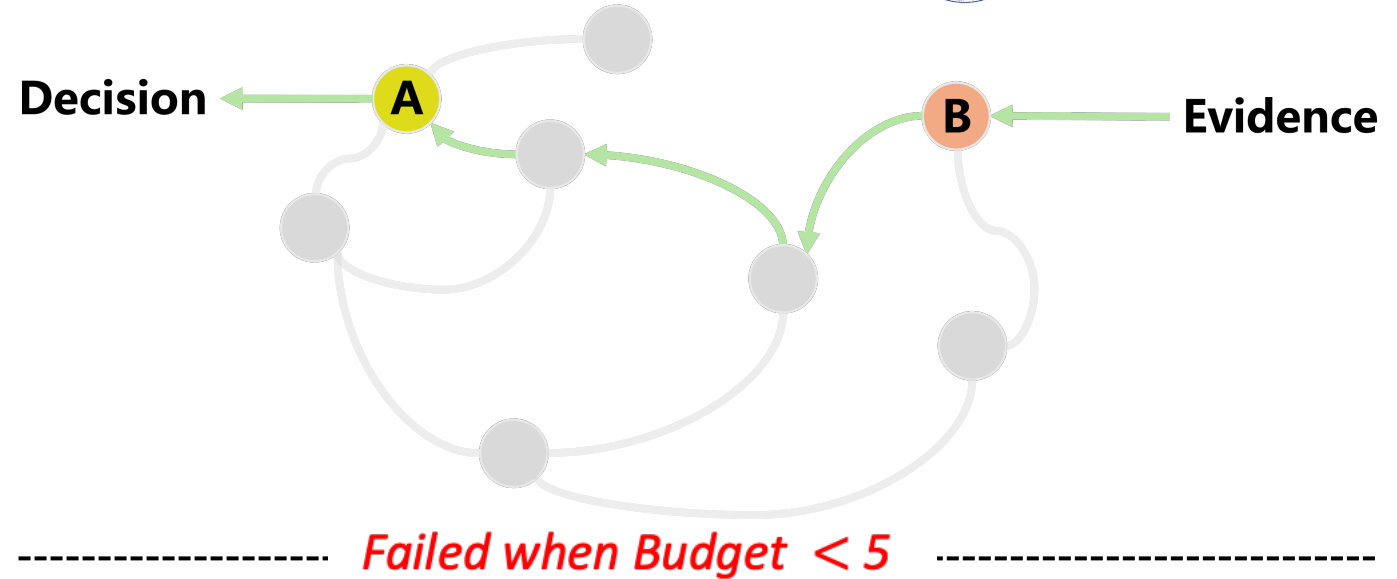
- *Information is distributed across different agents;*
- *communication delivers evidence to the decision points that need it.*



Split evidence → information routing

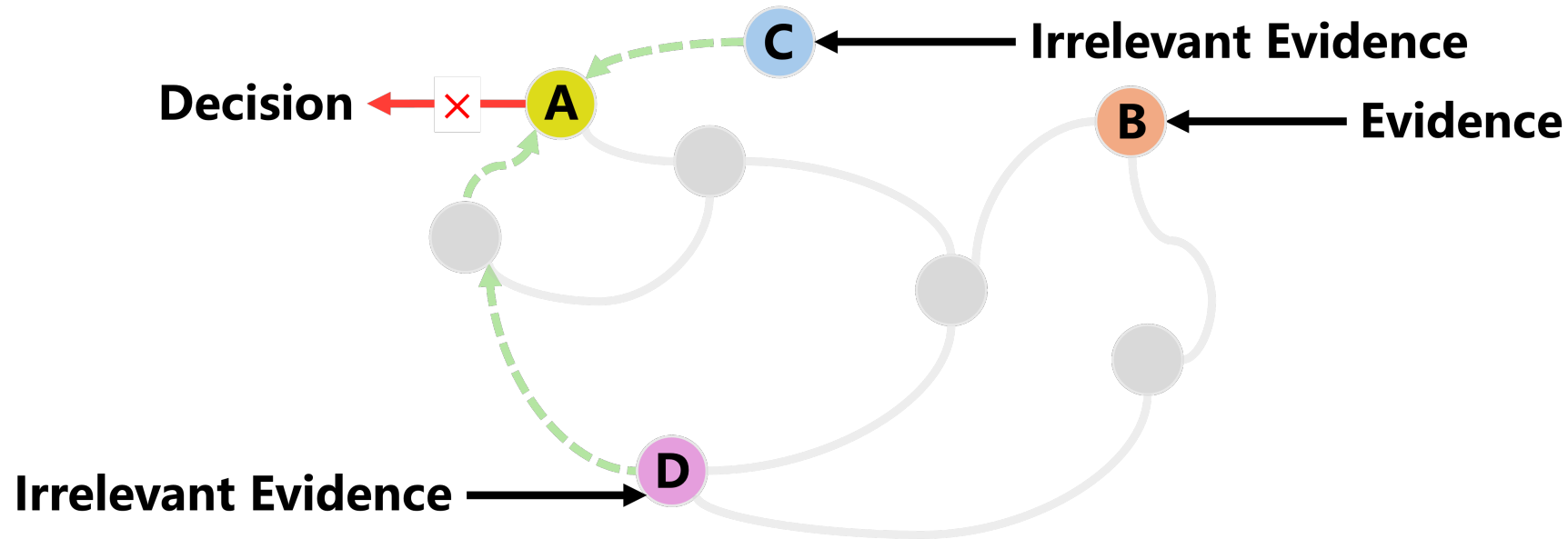
They can talk, but not freely.

- *Communication is costly* (bandwidth, tokens, turns, and topology...).
- *Scarce communication must be spent on the evidence that matters.*



More messages can still miss the demand.

- *Communication is only valuable when it targets unresolved demand;*
- *otherwise, it merely consumes budget.*



More communication $\neq$ better routing

From messages to evidence routing.

SBER rewrites

Message exchange

- *Ask what to say*
- *Message-first*
- *Weak link to decision demand*

Evidence routing

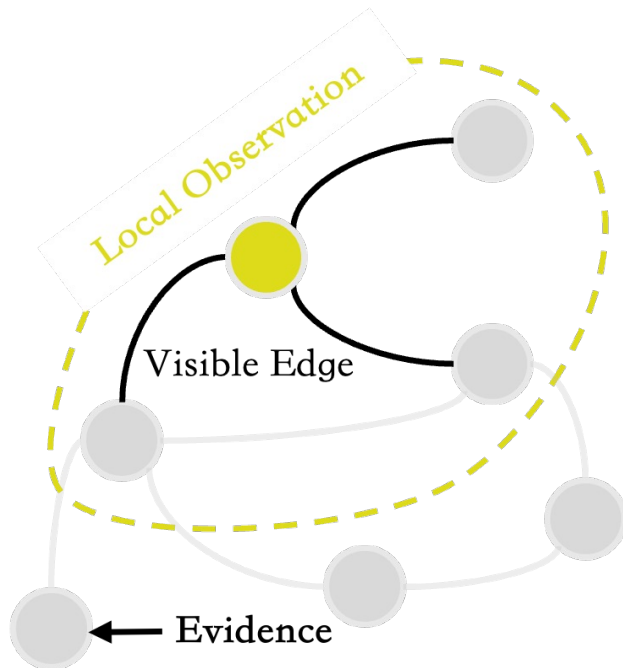
- *Ask what is missing*
- *Demand-first*
- *routes evidence to unresolved decision*



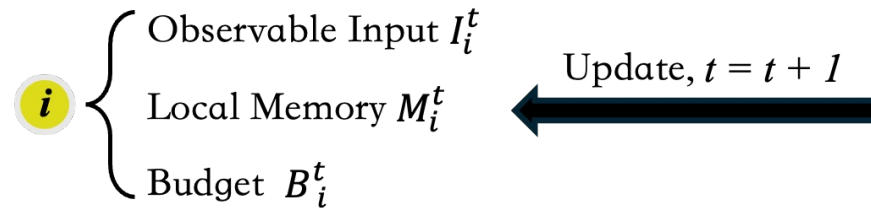
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SBER Overview

Budgeted Communication Graph



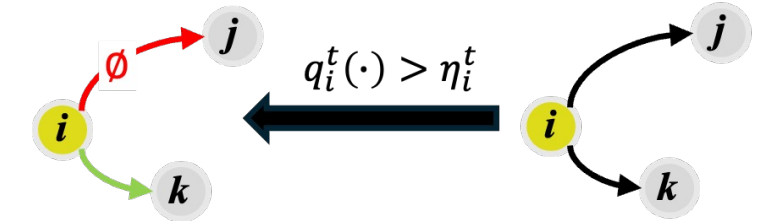
Setting of Agent i at t



Stage 1: Unsolved Demands Construction

Slot Construction at t	p	e
Missing evidence	0.8	0.5
Answer flip risk	0.6	0.4
Missing confirmation	0.7	0.5

Stage 3: Value Scoring & Decision



Stage 2: Candidate Generation

Candidate Generation	Budget
(Ask, Agent j , u_1)	✓
(Ask, Agent j , u_2)	×
(Answer, Agent k , u_3)	✓

Demand Slots

slot = (type, task object, priority, urgency)

missing_evidence

Key support for the current decision is absent from local view.

answer_flip_risk

A current judgment may change if another agent has different evidence.

missing_confirmation

A plan or answer needs confirmation before the team commits.



Action-Receiver-Slot Triple

act

ask / answer / tell

**receiver**

allowed neighbor

**slot**

unsolved demand

Free-from chat

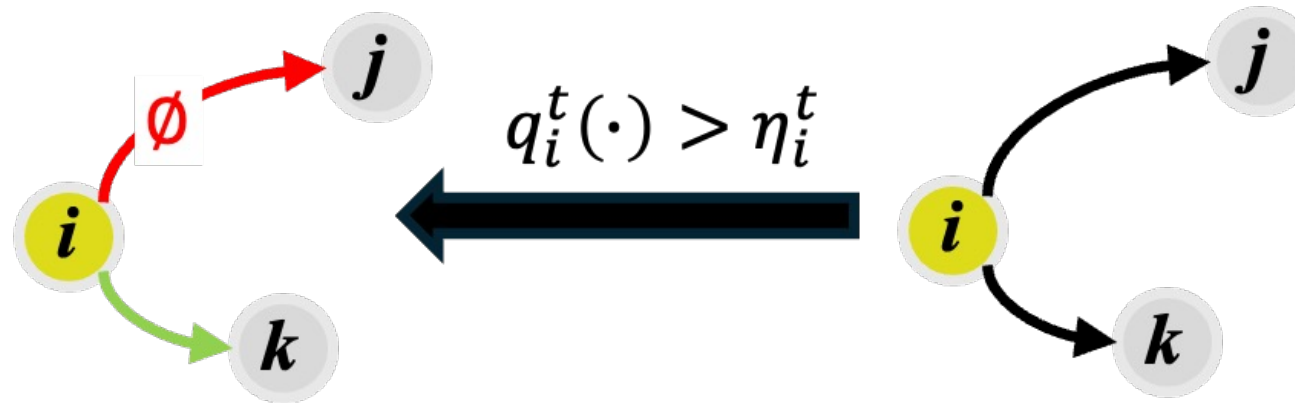
"I think maybe we should..."

Structured message

{ act, receiver, slot, cost }

Budget-Aware Routing

Stage 3: Value Scoring & Decision



score = value(slot, act, receiver) – cost pressure

send if score > adaptive threshold



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Research Questions

We evaluate whether demand-aware routing works, when it works, and why.

- *RQ1 : Does SBER improve coordination?*
- *RQ2 : Does it work under changing budgets?*
- *RQ3 : What drives the gains?*
- *RQ4 : Structure or LLM chat?*
- *RQ5 : How cautious should communication be?*

RQ1 | Does routing help?

Table 1. RQ1 headline results on InformativeBench. Entries are mean $\pm$ std over three seeds.

Method	Precision	F1-score	IoU	Cost
<i>Schedule-Medium</i>				
InfoNav	0.538 $\pm$ 0.021	0.452 $\pm$ 0.018	0.292 $\pm$ 0.015	979.120 $\pm$ 12.403
InfoNav w/o Rec.	0.507 $\pm$ 0.017	0.392 $\pm$ 0.014	0.244 $\pm$ 0.011	842.770 $\pm$ 9.826
InfoNav w/o Mem.	0.462 $\pm$ 0.015	0.347 $\pm$ 0.013	0.210 $\pm$ 0.010	978.430 $\pm$ 10.517
TarMAC	0.726 $\pm$ 0.012	0.583 $\pm$ 0.016	0.418 $\pm$ 0.015	661.420 $\pm$ 8.137
SBER	0.922 $\pm$ 0.007	0.842 $\pm$ 0.008	0.727 $\pm$ 0.010	861.720 $\pm$ 9.284
<i>Schedule-Hard</i>				
InfoNav	0.734 $\pm$ 0.027	0.772 $\pm$ 0.023	0.629 $\pm$ 0.027	973.450 $\pm$ 13.114
InfoNav w/o Rec.	0.708 $\pm$ 0.024	0.721 $\pm$ 0.020	0.564 $\pm$ 0.019	828.630 $\pm$ 9.441
InfoNav w/o Mem.	0.693 $\pm$ 0.021	0.704 $\pm$ 0.019	0.543 $\pm$ 0.017	972.110 $\pm$ 10.286
TarMAC	0.705 $\pm$ 0.018	0.716 $\pm$ 0.017	0.558 $\pm$ 0.018	655.880 $\pm$ 7.914
SBER	0.758 $\pm$ 0.012	0.754 $\pm$ 0.014	0.605 $\pm$ 0.016	968.460 $\pm$ 9.782

SBER improves coordination by routing evidence to the decisions that need it.

RQ1 | Does routing help?

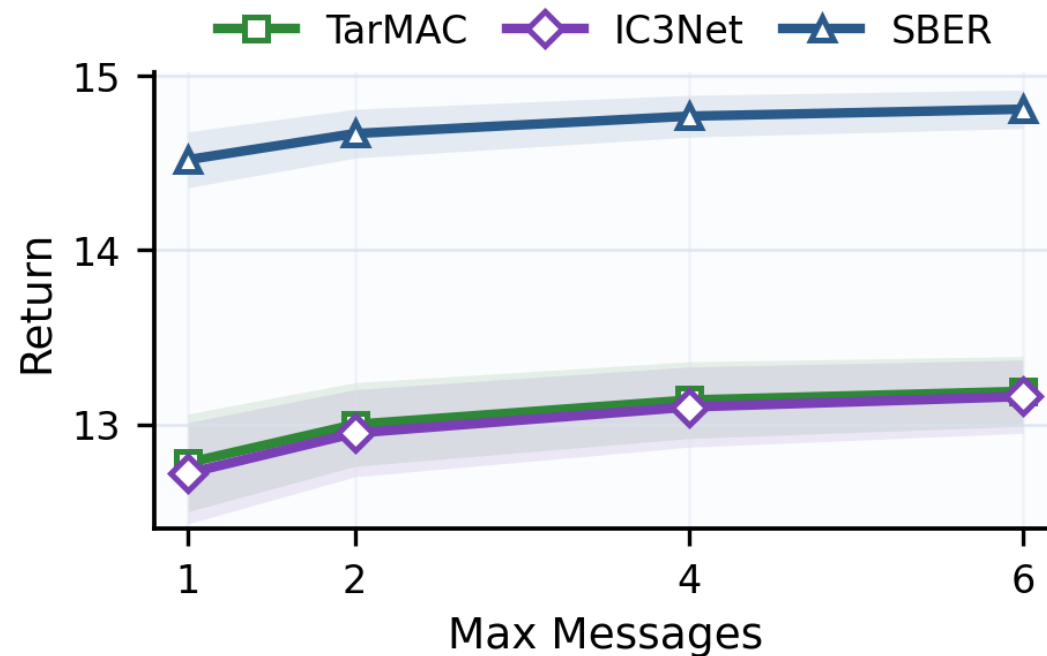
Table 2. RQ1 headline results on AdaSociety. Entries are mean $\pm$ std over three seeds.

Method	Return	Completion	Fairness
<i>Connection</i>			
Graph-Only	13.901 $\pm$ 0.231	0.928 $\pm$ 0.021	0.742 $\pm$ 0.024
Budget-Aware	13.948 $\pm$ 0.204	0.935 $\pm$ 0.018	0.756 $\pm$ 0.020
TarMAC	13.211 $\pm$ 0.257	0.936 $\pm$ 0.016	0.731 $\pm$ 0.022
IC3Net	13.138 $\pm$ 0.266	0.931 $\pm$ 0.019	0.724 $\pm$ 0.023
SBER	14.812 $\pm$ 0.143	0.996 $\pm$ 0.006	0.781 $\pm$ 0.014
<i>Overlapping Group</i>			
Graph-Only	13.947 $\pm$ 0.206	0.936 $\pm$ 0.018	0.884 $\pm$ 0.021
Budget-Aware	13.972 $\pm$ 0.191	0.939 $\pm$ 0.016	0.896 $\pm$ 0.018
TarMAC	13.158 $\pm$ 0.269	0.942 $\pm$ 0.015	0.861 $\pm$ 0.024
IC3Net	13.176 $\pm$ 0.252	0.938 $\pm$ 0.017	0.872 $\pm$ 0.022
SBER	14.847 $\pm$ 0.128	0.998 $\pm$ 0.004	0.918 $\pm$ 0.013
<i>Inequality</i>			
Graph-Only	13.884 $\pm$ 0.244	0.931 $\pm$ 0.023	0.286 $\pm$ 0.016
Budget-Aware	13.918 $\pm$ 0.219	0.934 $\pm$ 0.019	0.294 $\pm$ 0.014
TarMAC	13.205 $\pm$ 0.233	0.938 $\pm$ 0.017	0.301 $\pm$ 0.013
IC3Net	13.149 $\pm$ 0.241	0.934 $\pm$ 0.018	0.298 $\pm$ 0.014
SBER	14.833 $\pm$ 0.117	0.997 $\pm$ 0.005	0.318 $\pm$ 0.010

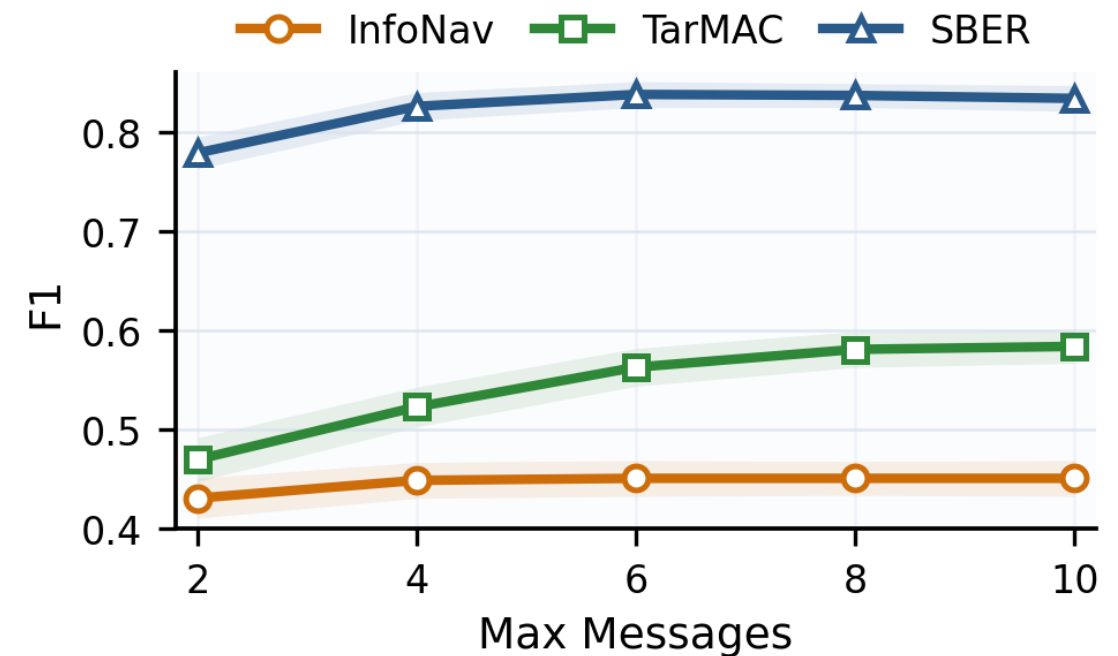
SBER improves coordination by routing evidence to the decisions that need it.

RQ2 | When budget changes

Inequality



Schedule-Medium



SBER is most valuable when communication budget is scarce.

RQ3 | What makes it work?

Table 3. RQ3 ablation study on the `private_information` protocol. Results are reported on `Schedule-Medium`. Entries are mean $\pm$ std over three seeds. Δ denotes the change relative to the full SBER model.

Method	Precision		F1		IoU	
	Value	Δ	Value	Δ	Value	Δ
SBER (full)	0.922 $\pm$ 0.007	0.000	0.842 $\pm$ 0.008	0.000	0.727 $\pm$ 0.010	0.000
w/o slots	0.698 $\pm$ 0.013	-0.224	0.532 $\pm$ 0.015	-0.310	0.362 $\pm$ 0.014	-0.365
w/o acts	0.556 $\pm$ 0.017	-0.366	0.454 $\pm$ 0.018	-0.388	0.294 $\pm$ 0.016	-0.433
w/o budget	0.578 $\pm$ 0.015	-0.344	0.476 $\pm$ 0.017	-0.366	0.312 $\pm$ 0.015	-0.415

Demand slots, communication acts, and budget control each contribute to the gains.



RQ4 | Structure vs chat

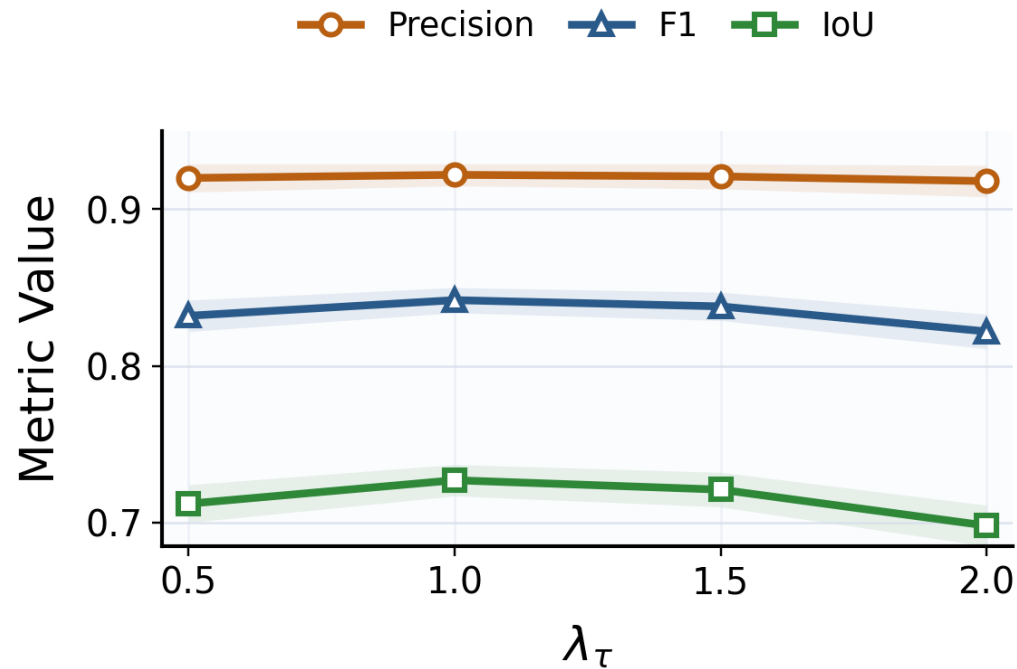
Table 4. RQ4 comparison between explicit protocol control and fully LLM-controlled communication on focused private and public settings. Entries are mean $\pm$ std over three seeds.

Private setting: Schedule-Medium			
Method	Precision	F1-score	IoU
SBER	0.922 $\pm$ 0.007	0.842 $\pm$ 0.008	0.727 $\pm$ 0.010
LLM-C	0.756 $\pm$ 0.014	0.684 $\pm$ 0.017	0.520 $\pm$ 0.018
TarMAC	0.726 $\pm$ 0.012	0.583 $\pm$ 0.016	0.418 $\pm$ 0.015
Public setting: Inequality			
Method	Return	Fairness	Cost
SBER	14.833 $\pm$ 0.117	0.318 $\pm$ 0.010	190.500 $\pm$ 5.214
LLM-C	13.800 $\pm$ 0.206	0.309 $\pm$ 0.014	212.000 $\pm$ 7.381
TarMAC	13.205 $\pm$ 0.233	0.301 $\pm$ 0.013	200.000 $\pm$ 6.148

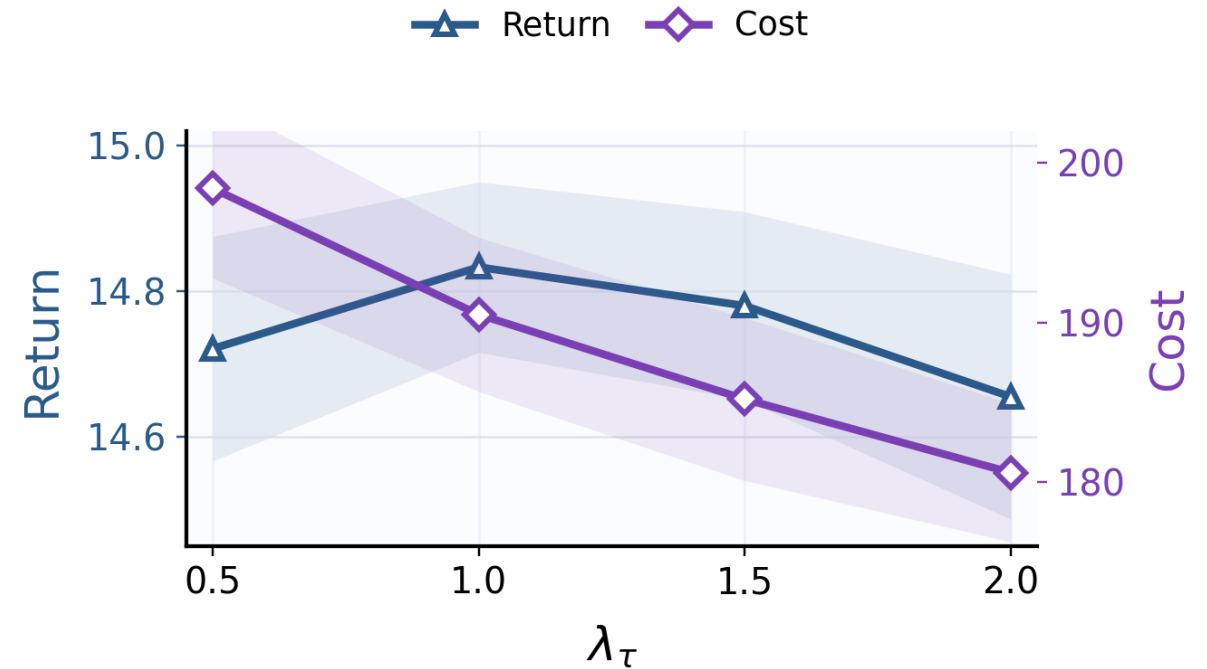
Structured protocol control is more reliable than unconstrained LLM chatting under constraints.

RQ5 | How cautious to be?

Schedule-Medium



Inequality



The send-or-silence gate works best with calibrated communication conservatism.



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Conclusion

Contribution

- *Demand-first communication;*
- *Structured routing control;*
- *Budget-aware allocation.*

***Every message should answer:
what evidence, to whom, for which demand?***



Thanks

Yan Li

liyanly@bupt.edu.cn